

GLOBAL CLOUD FRACTION ANALYSIS

Harmonization of ISCCP, CERES and CMIP6 Climate Models

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Tools Used: Python | Xarray | Pandas | NumPy | Matplotlib | Scikit-learn | Jupyter Notebook

1. Objective

Project Aim

The objective of this project is to evaluate the performance of CMIP6 climate models by comparing them with satellite-based cloud fraction observations.

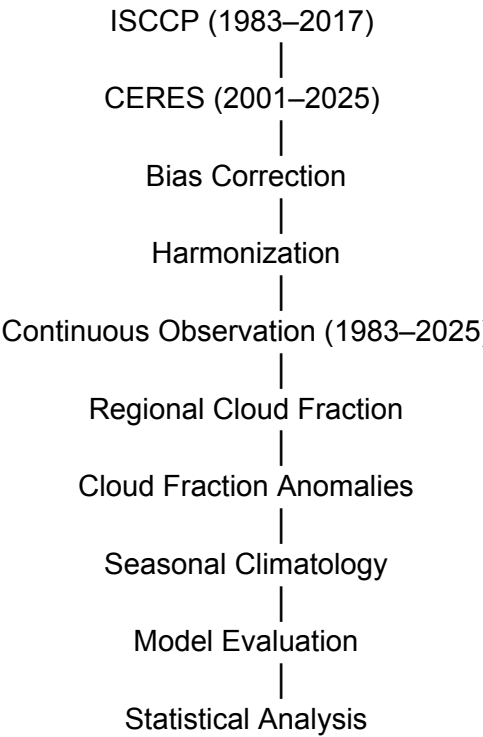
Specific Objectives

- Harmonize ISCCP (1983–2017) and CERES (2001–2025) satellite datasets.
- Create a continuous global cloud fraction observational record.
- Compare observations with three CMIP6 climate models:
 - UKESM1-0-LL
 - IPSL-CM6A-LR
 - MRI-ESM2-0
- Analyze cloud fraction across different latitude bands.
- Compute cloud fraction anomalies.
- Study seasonal climatology.
- Evaluate model performance using statistical metrics.

2. Datasets

Dataset	Variable	Time Period
ISCCP HGM	Cloud Fraction	1983–2017
CERES SYN1deg Ed4.2	Cloud Area Fraction	2001–2025
UKESM1-0-LL	Cloud Fraction (clt)	1983–2025
IPSL-CM6A-LR	Cloud Fraction (clt)	1983–2025
MRI-ESM2-0	Cloud Fraction (clt)	1983–2025

3. Workflow



4. Study Regions

Cloud fraction was evaluated over five geographical regions.

Region	Latitude Range
Global	Entire Earth
Tropics	30°S – 30°N
Mid-latitudes	30° – 60°
Polar Regions	60 – 90°

5. Harmonization Method

Since ISCCP and CERES cover different periods, a harmonized observational dataset was created.

Procedure

- ISCCP data were used from **1983–2000**.
- CERES data were used from **2001–2025**.
- The overlapping period (2001–2017) was used to estimate the mean bias.
- A bias correction was applied to ISCCP.
- The corrected ISCCP series was merged with CERES to obtain a continuous observational record (1983–2025).

6. Global and Regional Cloud Fraction

Include the following figures:

- Global Cloud Fraction
- Tropical Cloud Fraction
- Subtropical Cloud Fraction
- Mid-latitude Cloud Fraction
- Polar Cloud Fraction

Figure Caption Example:

Comparison of harmonized satellite observations (ISCCP + CERES) with CMIP6 climate models for global cloud fraction from 1983–2025.

7. Cloud Fraction Anomalies

Insert:

- Global Anomalies
- Tropics
- Subtropics
- Mid-latitudes
- Polar

Method

Monthly anomalies were calculated relative to the climatological monthly mean. A 12-month running mean was applied to emphasize long-term variability.

8. Seasonal Climatology

Include:

- Global Seasonal Climatology
- Tropical Seasonal Climatology
- Subtropical Seasonal Climatology
- Mid-latitude Seasonal Climatology
- Polar Seasonal Climatology

Description

Monthly climatology (January–December) was calculated to evaluate the seasonal cycle of cloud fraction and compare it with CMIP6 models.

9. Bias Analysis

Insert:

- Global Bias
- Tropical Bias
- Subtropical Bias
- Mid-latitude Bias
- Polar Bias

Definition

Bias = Model - Observation

Positive values indicate model overestimation, while negative values indicate underestimation of cloud fraction.

Statistical Evaluation

The following statistical metrics were used to assess model performance:

- Correlation Coefficient (R)
- Root Mean Square Error (RMSE)
- Bias
- Standard Deviation

Region	Model	Correlation (R)	RMSE	Std. Deviation	Bias
Global	UKESM1-0-LL	0.414	1.889	1.545	-0.152
Global	IPSL-CM6A-LR	0.184	7.003	1.540	-6.646
Global	MRI-ESM2-0	0.155	6.580	1.018	-6.267
Tropics	UKESM1-0-LL	0.346	4.834	1.428	-4.038
Tropics	IPSL-CM6A-LR	0.103	12.386	1.801	-11.976
Tropics	MRI-ESM2-0	0.068	7.091	1.181	-6.447
Mid-Latitudes	UKESM1-0-LL	0.610	2.954	3.511	-0.986
Mid-Latitudes	IPSL-CM6A-LR	0.510	6.207	4.553	-4.815
Mid-Latitudes	MRI-ESM2-0	0.565	7.479	2.335	-7.168
Polar	UKESM1-0-LL	0.509	28.301	3.907	27.796
Polar	IPSL-CM6A-LR	0.266	17.583	5.924	15.998
Polar	MRI-ESM2-0	0.471	7.293	5.535	4.128

Results Discussion

Global Region

UKESM1-0-LL demonstrates the best overall performance, with the highest correlation (0.414), the lowest RMSE (1.889), and the smallest bias (-0.152). In contrast, IPSL-CM6A-LR and MRI-ESM2-0 show larger negative biases, indicating a tendency to underestimate global cloud fraction.

Tropical Region

Model performance decreases in the tropics. UKESM1-0-LL again performs best, while IPSL-CM6A-LR exhibits the largest RMSE and strongest negative bias. MRI-ESM2-0 performs better than IPSL but still underestimates cloud fraction.

Mid-Latitude Region

The highest model agreement is observed in the mid-latitudes. UKESM1-0-LL achieves the strongest correlation (0.610) and the lowest RMSE (2.954), indicating that it captures cloud

variability more accurately than the other models. MRI-ESM2-0 has a relatively high correlation but also a larger negative bias.

Polar Region

The polar region shows the greatest differences among the models. Although UKESM1-0-LL has the highest correlation (0.509), it substantially overestimates cloud fraction, resulting in the highest RMSE and bias. MRI-ESM2-0 provides the lowest RMSE (7.293) and the smallest positive bias (4.128), making it the most reliable model for polar cloud fraction in this analysis

Overall Model Performance

	Model	Summary
1	UKESM 1-0-LL	Best overall performance across Global, Tropical, and Mid-Latitude regions, with consistently higher correlations and lower errors.
2	MRI-ES M2-0	Best performance in the Polar region due to its lower RMSE and smaller bias, though it underestimates cloud fraction elsewhere.
3	IPSL-C M6A-L R	Generally exhibits larger RMSE values and stronger biases, indicating comparatively weaker agreement with observations.

Key Findings

The major findings of this study are summarized below:

- A continuous cloud fraction observational dataset (1983–2025) was successfully developed by harmonizing ISCCP HGM and CERES satellite observations.
- The harmonized dataset enabled consistent evaluation of cloud fraction across four latitude regions: Global, Tropics, Mid-Latitudes, and Polar.
- Among the three CMIP6 models, **UKESM1-0-LL** showed the best overall agreement with observations in the Global, Tropical, and Mid-Latitude regions, achieving higher correlation coefficients and lower error statistics.
- **MRI-ESM2-0** demonstrated the best performance in the Polar region, with the lowest RMSE and smallest bias.
- **IPSL-CM6A-LR** generally exhibited larger RMSE values and stronger biases, indicating comparatively lower agreement with the observational dataset.
- The statistical evaluation highlighted regional differences in model performance, emphasizing the importance of evaluating climate models across different latitude bands.
- The complete workflow demonstrates the application of Python-based scientific computing for climate data harmonization, visualization, and model validation.

Technical Skills Demonstrated

Programming & Data Processing

- Python
- Jupyter Notebook
- Xarray
- Pandas
- NumPy

Climate Data Analysis

- NetCDF Data Processing
- Satellite Data Harmonization
- CMIP6 Climate Model Evaluation
- Global and Regional Cloud Fraction Analysis

Statistical Analysis

- Correlation Analysis
- Root Mean Square Error (RMSE)
- Bias Analysis
- Standard Deviation
- Time-Series Analysis
- Monthly Climatology

- Anomaly Analysis

Data Visualization

- Matplotlib
- Scientific Figure Preparation
- Comparative Time-Series Visualization
- Climate Model Performance Assessment

Conclusion

This project developed a comprehensive workflow for evaluating cloud fraction simulations from CMIP6 climate models using satellite observations. By harmonizing ISCCP HGM and CERES datasets, a continuous observational record spanning **1983–2025** was established for comparison with model simulations. Cloud fraction was analyzed at both global and regional scales, and model performance was assessed using statistical metrics including Correlation (R), RMSE, Standard Deviation, and Bias.

The results indicate that **UKESM1-0-LL** provides the most consistent agreement with observations over the Global, Tropical, and Mid-Latitude regions, while **MRI-ESM2-0** performs best in the Polar region. This study demonstrates the value of combining multiple satellite datasets with climate model outputs to evaluate cloud representation and highlights the importance of regional assessments in climate model validation.